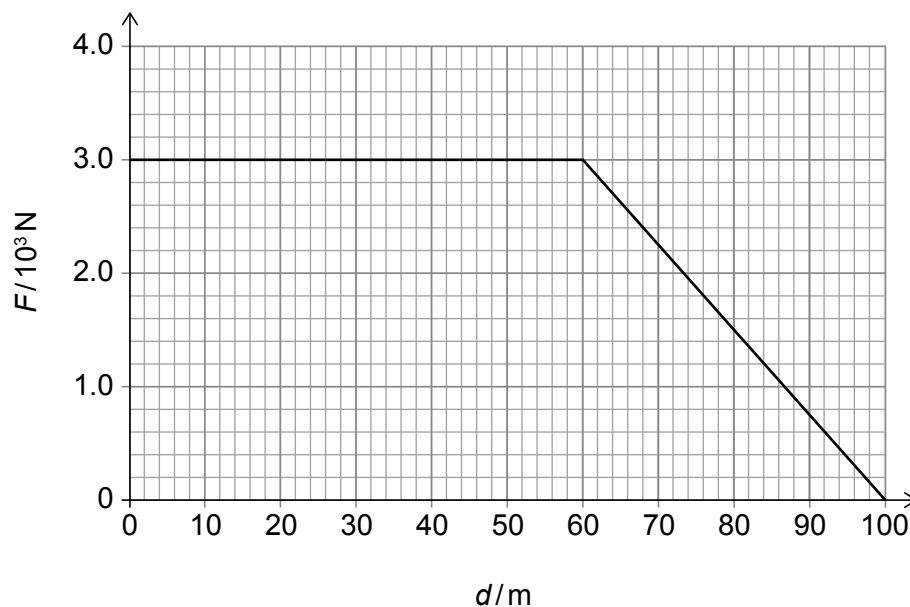


PAPER 2 / QUESTION 21

30 April 2025 / TZ1 (file label) / HL Paper 2 / original question 1 / 2225-6709

- 21.** (a) A car of mass 1600 kg accelerates from rest.

The graph shows how the resultant force F acting in the direction of motion of the car varies with the distance d travelled by the car.



- (i) State what is represented by the area under the graph.

[1]

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- (ii) Calculate the final speed of the car.

[2]

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(This question continues on the following page)

PAPER 2 / QUESTION 21

30 April 2025 / TZ1 (file label) / HL Paper 2 / original question 1 / 2225-6709

(Question 21 continued)

A different car travels on a horizontal road at a constant speed of 45 m s^{-1} . The engine of the car develops a power of 140 kW . The resistive force F_d acting on the car is given by

$$F_d = cv^2$$

where v is the speed of the car and c is a constant.

(b) Determine c . State the fundamental SI unit for your answer.

[3]

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